

Capstone Project Final Report

Sales Activities Analytics:

Understanding Sales Journeys, Follow-ups, and Deal Outcomes

Group 1 The Pioneers

Leader: Jason Wong

Member: Ben Tsui, Ian Wong, Janet Tong, John Ho, Nina Woo

Project Objectives

The primary objective of this analysis is to evaluate sales performance and optimize lead conversion strategies by leveraging 31,874 historical activity logs from a Hong Kong recruitment and job posting services provider.

The analysis focuses on three core tasks. First, it aims to standardize action item categories by extracting insights from unstructured text descriptions, turning individual messages into structured activity types. Second, to enable tracking across multiple interactions, the project involves assigning a unique Company ID to each organization. By ensuring that multiple rows belonging to the same company share a consistent identifier, the analysis can link disparate log entries into a chronological sequence of events. Finally, the report seeks to map the sales journey to determine the average duration and number of touchpoints required to close a deal. By identifying the frequency of interactions before a deal is abandoned, these insights will define a data-driven threshold for "giving up", helping the team refine resource allocation and improve sales efficiency.

Dataset Description

I. Introduction to the dataset

The dataset utilized for this analysis consists of 31,874 historical sales activity logs generously provided by CPJobs. These records offer a comprehensive view of the interaction history between the sales team and various organizations, encompassing both potential and existing clients, offering a realistic view of recruitment sales engagement from 2023 to 2026.

II. Dataset characteristics

The data is structured into 6 primary features that provide the foundation for our behavioral analysis:

Column	Description
CompanyName	The formal name of the organization
Industry	The business sector or category the company belongs to
LogType	A categorical field defining the status or outcome of an activity. Key categories include: Signed, Follow-up, Proposed, Unreached and Rejected
Log Date	The specific date and time of the activity
message	A free-text column recording the detailed conversation or notes from each interaction
DealAmount	The financial value associated with the transaction or account

III. Data preprocessing steps

To ensure the integrity and consistency of the analysis, several data cleaning and standardization steps were performed:

1. Handling Missing Values:

- a. **Industry:** Missing entries in the Industry column were categorized as 'Unknown' after consultation with the client to maintain the completeness of the dataset.
- b. **Deal Amount:** Following client feedback, since many are missing until closure and client suggested we can skip monetary details — we shifted focus to a simpler, more reliable outcome: whether the deal was Signed or not. Missing financial values set to 0.

2. Text Standardization & Abbreviation Expansion:

The Message column contained various domain-specific shorthand and typos that could hinder categorization. To address this, an Abbreviation Expansion Dictionary was developed to map and interpret shorthand (e.g., expanding “LVM” to “Leave Voice Message”). This ensured that the text analysis model could accurately interpret the sales activities.

3. Contextual Data Cleaning:

Targeted logic was applied to the Message field to improve interpretability and consistency:

- a. **Value-Action Integration:** When a message was less than five words long and conformed to the rules, the message contained specific terms like “10T”, the text was standardized to “signed 10T” to capture both the outcome and the detail in a single feature.
- b. **Message Imputation:** In cases where the Message field was empty, the entry was populated with its corresponding LogType to ensure every row had a descriptive label for categorization.
- c. **Message Optimisation:** The messages typically contained a meaningless Chinese word, an abbreviation of a Chinese name. All of these characters were removed to facilitate efficient machine learning training.

Task A - Categorization

From the free text description, define different categories for action items

The primary objective of Task A was to transform 31,874 unstructured free-text messages into standardized action categories. To achieve the highest level of analytical precision, the project followed an iterative methodology, transitioning from fully automated unsupervised learning models to a refined hybrid approach.

I. Methodology: Comparative Approaches

1. Approach 1: Automated Topic Modeling and LLM Labeling

The initial phase utilized advanced NLP frameworks—including **BERTopic**, **FuzzyTM**, and **FASTopic**—integrated with **Large Language Models (Llama 3.1 and 3.2)** for automated cluster identification. Despite the technical sophistication of these models, they encountered significant structural limitations:

- a. **Data Dispersion and Outliers:** **BERTopic** (*Appendix B, Figure B1*) identified 10 clusters but resulted in over **5,500 outliers** and excessive data collapse, where more than **14,000 rows** were inappropriately consolidated.
- b. **Contextual Inaccuracy:** While **FuzzyTM** (*Appendix B, Figure B2*) allowed for overlapping clusters, the subsequent LLM-based mapping produced category labels that were contextually misaligned with the actual sales activities.
- c. **Extreme Data Concentration:** **FASTopic** (*Appendix B, Figure B3*) exhibited severe skewness; the top two categories, **"Refining Job Specifications"** (12,573 rows) and **"Verifying Candidate**

Qualifications" (12,365 rows), accounted for approximately **78% of the entire dataset**. This lack of granularity made it impossible to distinguish between nuanced sales stages.

- d. **Contextual Misalignment:** While the LLM generated professional labels, manual audits revealed that messages were often assigned based on surface-level keywords rather than the actual business intent.

2. Approach 2: Hybrid Clustering and Manual Refinement (Final Selection)

To resolve the inaccuracies of Approach 1, a **hybrid methodology** was adopted:

- a. **K-Means Clustering:** Employed to establish a mathematical baseline by grouping the 31,874 messages based on text similarity features.
- b. **Manual Audit via OpenRefine:** To ensure domain-specific accuracy, **OpenRefine** was used to conduct a detailed manual audit. This allowed for the correction of misclassified clusters and the merging of similar activities that automated models had previously split.

II. Final Outcome and Data Distribution

The transition to the Hybrid Methodology (Approach 2) effectively resolved the data concentration issues identified in earlier automated trials. By moving away from the skewed distribution where 78% of the data was collapsed into two generic labels, the final classification achieved a more granular and analytically useful spread across 10 distinct action categories.

1. Data Distribution Highlights (Figure 1)

The final results for the 31,874 entries show a much more balanced and useful spread across the categories:

- a. **Primary Sales Activities:** The categories **"Job Related"** and **"Follow-up"** now make up about **49%** of all logs. This shows the main day-to-day work of the sales team without over-concentrating the data.
- b. **Customer Interaction:** Activities labeled as **"Communication"** and **"Services"** account for nearly **31%** of the records, reflecting steady engagement with both new prospects and existing clients.
- c. **Data Accuracy:** The number of unclassified rows (**"N.A."**) was reduced to only **0.18% (56 rows)**. This proves that the manual cleaning process captured almost every interaction that the automated models missed.

	Count	Percentage (%)
category		
job related	8866	27.82%
follow up	6822	21.40%
communication	5118	16.06%
services	4921	15.44%
unreached	2338	7.34%
rejected	2325	7.29%
signed	677	2.12%
financial related	483	1.52%
issue	268	0.84%
N.A.	56	0.18%

Figure 1 Action Categories from Hybrid Clustering (Approach 2 - Final Model)

Task B - Company Grouping

Grouping the data into a unique company and create unique company ID

The initial stage of the analysis required a structured approach to link 31,874 individual log entries to their respective organizations. Since each row represents a separate interaction with potentially inconsistent naming, Entity Resolution was performed using Python and the Pandas library to ensure data integrity.

I. Methodology: Entity Resolution and Unique Company ID Mapping

1. Data Normalization

To prevent duplicate entities caused by trailing spaces or formatting errors, the CompanyName column underwent a string stripping process (strip). This ensured that "Company A "(with space) and "Company A" were recognized as the same entity.

2. Unique Identifier Generation

After isolating the unique set of company names, a mapping dictionary was programmatically generated using a for loop and enumerate. Each unique company name was assigned a specific Numeric ID.

3. Data Mapping

Applied the dictionary to the entire dataset to create a new Company ID column. This linked multiple rows belonging to the same company to a single identifier.

II. Analytical Value:

This step was essential for converting a flat list of logs into a trackable history. It allowed our team to group all messages by their respective Company ID, providing the foundation for calculating the total duration and the number of follow-ups for each sales journey.

Task C - Sales Journey Insights

Find insights, define number of trials before giving up the deal, duration and number of tries before closing the deal

I. Methodology: Data Refinement and Metric Calibration

To ensure the integrity of this analysis, we addressed a primary challenge in our raw datasets: the absence of a distinct indicator to separate new sales follow-ups from post-sales administrative services.

1. Keyword-based Categorization

We performed a deep-dive analysis of the message content within every follow-up log. By identifying the 60 most frequent keywords, we bifurcated the logs into two segments:

- a. **Sales Attempts:** Logs containing intent-driven terms such as 'package', 'quote', 'offer', or 'need'.
- b. **Post-Sales/Admin:** Logs containing operational terms such as 'invoice', 'payment', or 'amend'.

2. Success Path Calibration

For analysis of how deals actually close, we focused exclusively on companies that signed deals. Each journey was tracked by a unique Company ID, allowing us to measure the number of attempts and duration required for closure.

3. The 180-Day Sales Cycle

We established a 180-day window to define each sales cycle. Any deal signed beyond this 180-day threshold is likely driven by the client's internal new requirements rather than our direct sales influence. This limit ensures our metrics accurately reflect our active influence on the deal.

4. Recovery definition

Recovery is defined as a company previously marked Rejected that later reached Signed status. After correction of tagging errors, we identified 20 verified recovery cases for detailed analysis.

II. The 90-Day Golden Window and ROI Disparity (Figure 2)

Focused on the 20 verified recoveries, conversion efficiency is dramatically higher in the first 90 days.

1. The High ROI Zone (Day 0-90)

Our analysis identifies a clear "**90-Day Golden Window**":

- 45% of total recoveries** (9 out of 20 cases) were achieved within the first 90 days.
- The **ROI Peak** (cumulative success rate) reaches its maximum efficiency at the 3-month mark (**0.50%**), with growth plateauing significantly thereafter.

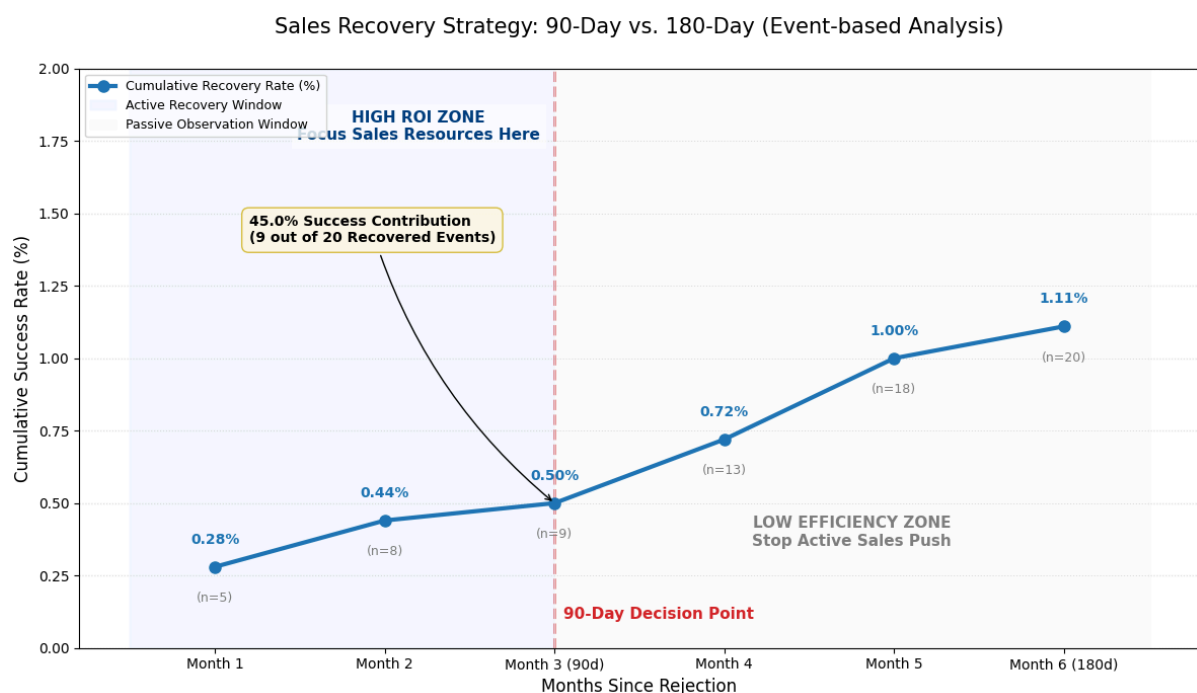


Figure 2 Sales Recovery Strategy: Cumulative Recovery Rate

2. Efficiency Gap Analysis (Figure 3)

There is a stark contrast in conversion efficiency between the early and late stages:

- Golden Window** (Day 0-90): We achieved a **5.77% conversion rate** (9 successes from 156 attempts).
 - Long-tail Period** (Day 91-180): Despite investing **384 sales attempts** (representing **71% of total effort**), we only secured 11 recoveries, resulting in a dropped conversion rate of **2.86%**
3. **Finding:** The early stage offers **2.02x higher ROI**. Sustaining high effort beyond 90 days does not justify the associated labor costs.

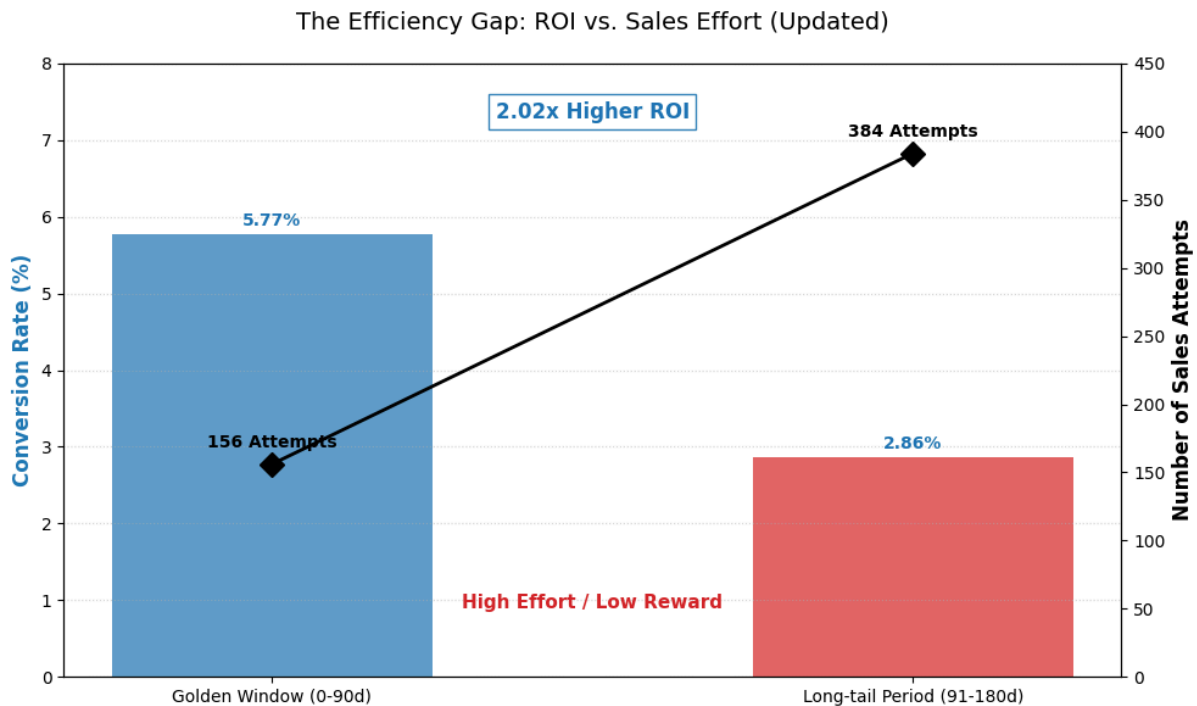


Figure 3 The Efficiency Gap: Conversion Rate vs. Sales Effort (0–90 vs. 91–180 Days)

III. The "Magic Number" of Attempts (Appendix C, Figure C1)

By analyzing the volume of effort required for a successful recovery, we discovered that "3" is the definitive threshold.

1. In the 90-day window, **100% of successful recoveries** were captured within **just 3 sales attempts**. Attempts beyond the 3rd produced zero additional gains. Even for the full 180-day sample, we reached a 90% success threshold by the 3rd attempt.
2. **Finding:** If a client intends to sign, they typically do so early in the follow-up process. Excessive persistence shows diminishing returns.

IV. Recovery Matrix and Strategic Summary (Appendix C, Figure C2)

The Heatmap crosses **Time** with **Sales Effort** to provide a strategic roadmap for resource allocation:

1. **The Low Effort "Sweet Spot":** In the High ROI Zone (Top Left), 45% of successes were achieved with minimal effort (0-3 tries). This confirms that in the first 90 days, **timing is a more powerful driver than persistence**.
2. **Passive Success:** Notably, **35% of all recoveries** (7 cases) occurred with **ZERO sales attempts**. This proves the impact of "Passive Success"—where simply staying top-of-mind is as effective as an active

push.

3. **The Effort Surge:** In the 91-180 day window, success shifts heavily toward the **"Standard Effort" (1-3 tries)** category. Recoveries in this late stage require more "heavy lifting" and active follow-up to finalize.

V. Cumulative Success Rate vs Sales Attempts (Figure 4)

1. Analysis shows that most deals are closed within 5–7 attempts, with the cumulative success rate plateauing after this point.
 - a. Success rate for signed deals showing rapid increase to ~96% by 7 attempts, proving that early engagement is the strongest driver of success.
 - b. Persistence beyond the 7th attempt yields minimal incremental gains, suggesting that resources should be reallocated rather than extended indefinitely.

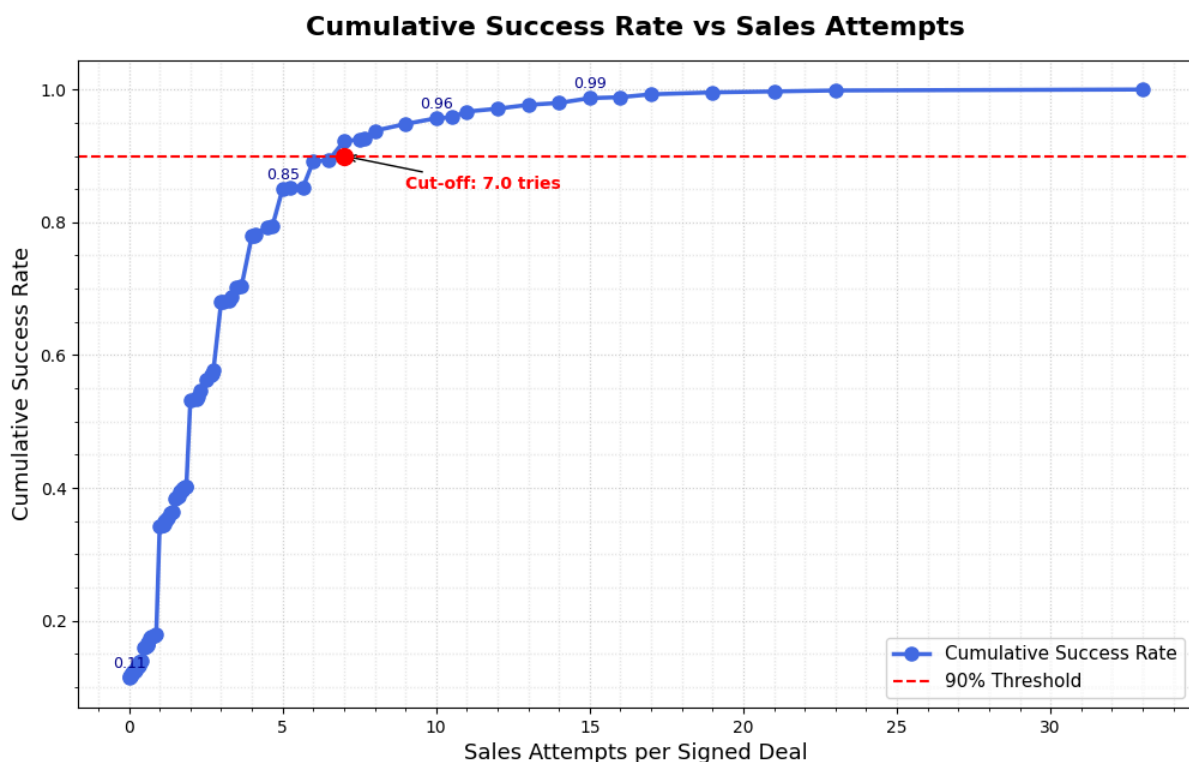


Figure 4 Cumulative Success Rate vs. Sales Attempts (Signed Deals Only)

2. **Analytical Insight:** This pattern confirms that quality outweighs quantity in follow-ups. Early-stage attempts must be carefully crafted and personalized, as they carry the highest probability of conversion. Beyond the 7th attempt, the diminishing returns highlight the need for a formal abandonment KPI to prevent wasted effort.

VI. Deal Success Distribution by Duration and Follow-Up Category (Figure 5)

1. Success outcomes vary significantly by deal duration:

- a. Short-term (0–10 days): Proactive sales pushes accelerate closures.
- b. Mid-term (10–30 days): Administrative and post-sales processes become critical for conversion.
- c. Long-term (60–180 days): Client-driven progress dominates, making heavy follow-up investment inefficient.

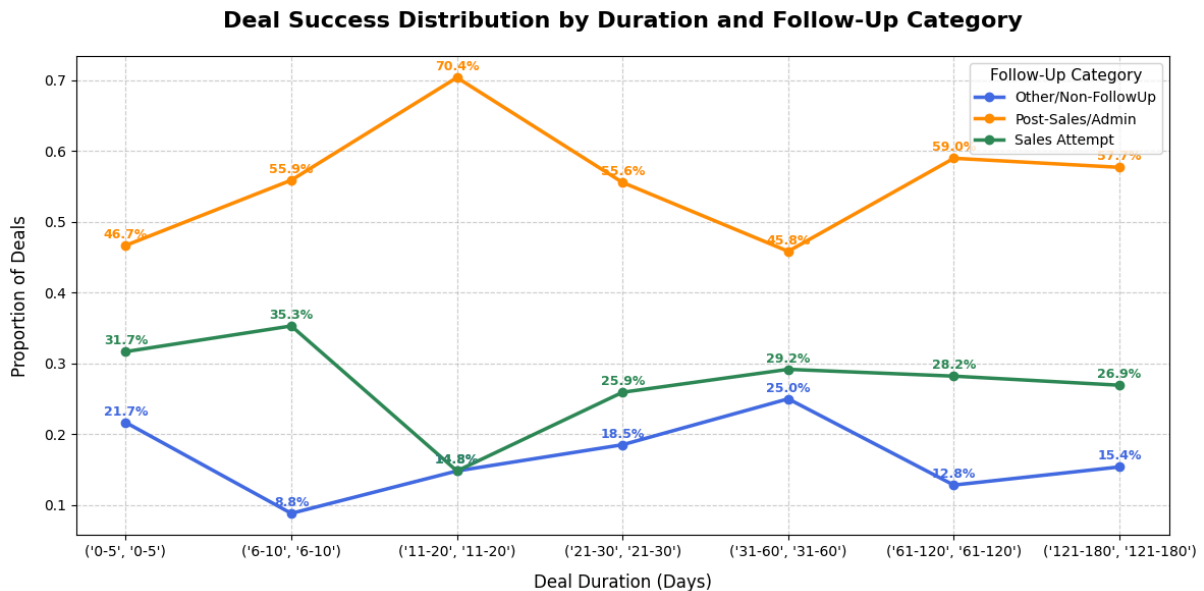


Figure 5 Deal Success Distribution by Duration and Follow-Up Category

2. **Analytical Insight:** The distribution shows that different stages require different resource strategies. Short-term wins depend on aggressive outreach, while mid-term deals highlight the importance of operational efficiency. Long-term deals, however, are largely outside direct sales influence, meaning teams should monitor rather than over-invest.

VII. Journey Sequences Outcome Distribution (Appendix C, Figure C3)

1. From the dataset of tracked journeys:
 - a. 8% of journeys ended in success.
 - b. 75.9% stalled without resolution, highlighting the need for stronger engagement and follow-up quality.
 - c. Successful paths consistently demonstrated faster closure cycles and stronger customer interaction.
2. **Analytical Insight:** The high incompleteness rate signals a major opportunity gap. By studying successful journeys, organizations can replicate winning behaviors (e.g., early personalization, clear value communication) and address drop-off points. This also suggests that objection handling and sustained engagement are critical levers for reducing failure rates.

VIII. Summary of Findings

The success analysis confirms that timing and quality of early attempts are decisive drivers of conversion. Sales teams should:

1. Prioritize effort within the first 90 days.
2. Set a reasonable abandonment point at 7 attempts.
3. Replicate winning early-stage behaviors to reduce the high rate of incomplete journeys.

IX. Strategic Implications

To maximize efficiency, organizations should front-load effort during the initial passive window, concentrating on the first 3–7 attempts where conversion likelihood is highest. Strengthening early engagement ensures momentum, while setting a disciplined threshold prevents wasted effort. By remaining highly selective in late-stage pursuits, resources are directed toward high-probability opportunities, improving overall conversion rates and reducing the cost of incomplete journeys.

Challenges and Limitations

I. Discussion of challenges faced during the project:

1. Short and mixed messages

Most messages were brief, full of abbreviations, or combined sales notes with routine admin updates. This reduced the effectiveness of fully automated text clustering and required substantial manual refinement.

2. Domain-specific abbreviations and phrasing variation

Salespeople used a lot of shorthand (e.g. “LVM”, “10T”, “f.u”, etc.) and inconsistent wording made semantic understanding difficult for standard NLP models without custom preprocessing and a dedicated abbreviation dictionary.

3. High outlier rates and fragmented topic structures

When using automatic clustering, a large number of messages were labeled as “outliers” or formed small, unclear groups. This needed several rounds of tuning and human judgment to produce meaningful categories.

4. Variable journey completeness

A large proportion of journeys never reached a clear outcome (Signed, Rejected, or Unreached), which introduced some uncertainty when calculating average success metrics.

II. Limitations of the approach or dataset

1. Noisy and inconsistent free-text messages

The free-text notes reflect real daily work but contain typos, abbreviations, and mixed admin topics. This naturally limits how accurately fully automatic analysis can be performed.

2. Missing metadata and financial data

Most records lack DealAmount, so we couldn't connect effort directly to revenue. There is also no data about which salesperson handled the lead, where the lead came from, or specific customer type — making it harder to explain why some deals needed many contacts while others closed quickly.

3. Selection bias from incomplete journeys

Because many conversations are still open or were never closed, the statistics on duration, number of follow-ups, and success patterns come mainly from cases that reached an end. This may not fully represent very long cycles or currently active opportunities.

4. Descriptive scope and model sensitivity

The work is descriptive rather than predictive. Time constraints prevented building more advanced models or running formal statistical tests.

III. Potential areas for future improvement

1. Better ways to clean abbreviated messages

Create stronger rules or use simple AI tools to automatically fix/expand notes and make them easier for computers to understand. This would reduce manual work and improve category accuracy.

2. Hybrid classification for better content separation

Combine unsupervised topic modeling with supervised or zero-shot LLM classification to better separate sales vs. administrative content and generate more interpretable per-message labels.

3. Enriched features and predictive modeling

If data access allows, incorporate additional company data (size, industry) or CRM metadata (salesperson, lead source) to enable more detailed analysis. Build lightweight predictive models to estimate deal-closure probability based on early journey signals.

4. Interactive delivery and monitoring

Convert the current Jupyter-based pipeline into a lightweight dashboard for ongoing client exploration of topics, journeys, and performance metrics.

Future Work

I. Description of any additional ideas or approaches that were not implemented

1. Enrich very short messages automatically
Enrich short messages (e.g., "1 URL", "1 post", "LVM") by inferring verbs/context via simple rules or light AI prompts to improve text clustering quality.
2. Build real-time automatic labeling
Develop an automated real-time pipeline to categorize new messages instantly.
3. Test more modern text understanding models
Conduct broader benchmarking of short-text models (e.g., Sentence-BERT, E5, LLM zero-shot classification) to find the best fit for domain-specific CRM notes.
4. Make a clean summary table and nice visuals
Build one clean company-level analytical table (columns: CompanyId, Industry, First/Last contact, Total touches, Follow-up count, Outcome, Days to outcome, Top activities, Failure patterns) plus interactive charts (paths to win/lose, funnel, touch-count bars). This would make insights much easier to share and use daily.

II. Reasons for not implementing these ideas

1. Time constraints within the 1-month Capstone timeline limited focus to core deliverables (stable categories, journey metrics, key insights). More complex automation, advanced model comparison, and polished dashboard development required additional iteration and resources beyond the project scope. Priority was given to interpretable, business-actionable results using hybrid approaches rather than production prototypes.

Conclusion

I. Summary of the project's objectives and achievements

This project successfully met its core objectives by transforming 31,874 raw, unstructured sales activity logs from a Hong Kong recruitment services provider into structured, actionable business insights.

Key achievements include:

- a. Defining 10 stable and interpretable action-item categories from free-text messages using a hybrid clustering approach.
- b. Constructing complete sales journeys at the company level through unique Company ID assignment and temporal sequencing.
- c. Quantifying critical sales performance metrics, including average deal duration, typical number of touchpoints for success, the “power of 3” attempts (90% of quick recoveries within 3 attempts), and the 90-day golden window for highest conversion efficiency.

These outcomes were delivered through proficient application of Python (Pandas, NumPy), exploratory data analysis, unsupervised text clustering techniques, and thoughtful business-oriented interpretation — demonstrating originality in tackling real-world CRM short-text challenges.

II. Key takeaways and insights gained

The analysis yielded several high-impact, data-driven recommendations for the sales team:

- a. Concentrate resources and follow-up efforts heavily in the **first 7 attempts** and especially within the **first 90 days** — beyond these points, marginal returns diminish rapidly.
- b. Automate or streamline routine follow-up activities (21% of all logs) to free up time for higher-value engagements.
- c. Route service-related requests (15% of interactions) to dedicated support queues to prevent them from cluttering the sales pipeline.
- d. Prioritize early-stage acceleration: proactive sales attempts in the 0–10 day window show the strongest association with eventual success.

In summary, this project gives clear, evidence-based ways to focus sales effort where it matters most: early, focused, and high-quality engagement. The Pioneers team has shown how targeted data science techniques can uncover practical ways to improve conversion rates and sales efficiency in a competitive recruitment market.

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Distribution of Work

- Jason Wong – Project Lead & Quality Assurance
 - Coordinated with client and instructor, managed timeline and scope.
 - Reviewed deliverables (slides, code, report) and ensured consistency across all tasks.
- Janet Tong – Analytics & Quality Assurance
 - Designed overall analysis plan and key KPIs (conversion, attempts, duration).
 - Validated metrics, checked numbers in charts/tables, and refined business wording of insights.
- Ian Wong – Data Analysis & Processing
 - Led data cleansing and preprocessing (abbreviation mapping, industry standardization, DealAmount handling).
 - Implemented company-level grouping and creation of unique Company IDs.
- Nina Woo – Data Analysis & Processing
 - Performed exploratory data analysis and visualized journey patterns and distributions.

- Supported preparation of intermediate datasets for modeling and Task C calculations.
- Ben Tsui – Machine Learning & AI
 - Led the “Define cluster and refine results” work using tools like OpenRefine and clustering methods to group similar messages.
 - Helped validate and clean the final category assignments to ensure they were consistent and business-readable.
- John Ho – Machine Learning & AI
 - Led the “AI generation of categories” work, experimenting with LLMs and topic-modeling approaches (e.g., BERTopic, FuzzyTM, FASTopic).
 - Designed and tested prompts and workflows for using AI to propose category labels and evaluated their effectiveness.

Appendix A

Figure A1

Setup Instructions from Project README

 README

Quick Start – One-Click Setup

We created a **dedicated setup notebook** that installs everything you need automatically.

Step 0: Open and run the setup notebook first

1. Open → `notebooks/00-setup-and-install.ipynb`
2. Run **all cells** (especially the first two big ones)
3. It will:
 - Detect missing packages
 - Install them quietly (`bertopic` , `sentence-transformers` , `hdbscan` , `umap-learn` , `ollama` , etc.)
 - Give clear error messages if anything goes wrong
4. **Restart the kernel** when it asks (very important on Colab/Jupyter)
5. Run the verification cell → you should see green ✓ for all packages

→ After this step, **all other notebooks should run without installation problems.**

Recommended Execution Order

1. **Setup**
 - `00-setup-and-install.ipynb` (run once per environment)

Note. "Quick Start" section from README.md, detailing the mandatory first step of running the automated installation notebook. Critical for reproducibility of all topic modeling and classification experiments.

Figure A2

Excerpt of the Abbreviation Expansion Dictionary used in Text Preprocessing

```
Datasets > {} abbr_lookup.json > ...  
1  {  
2    "&": "and",  
3    "1111": "seasonal",  
4    "a/c": "account",  
5    "ac": "account",  
6    "acc": "account",  
7    "acct": "account",  
8    "ad": "advertisement",  
9    "ads": "advertisement",  
10   "ans": "answer",  
11   "atm": "at the moment",  
12   "att": "attachment",  
13   "br": "business registration",  
14   "cfm": "confirm",  
15   "CNY": "seasonal",  
16   "cp": "cpjob",  
17   "cpt": "cptalent",
```

Note. Partial view of `abbr_lookup.json` showing mappings of common sales-log abbreviations and shorthand (e.g., "f.u." → "follow up", "CNY" → "seasonal") to their expanded forms. The full dictionary contains 51 entries and was applied during data cleaning to improve text quality for topic modeling and classification. Created in `abbr_dict_builder.ipynb`.

Appendix B

Figure B1

Distribution of Topic-Action Labels Generated by BERTopic (Approach 1a)

```
1 df['topic_action_label'].value_counts()
```

topic_action_label	
Follow-up Email Attempts Necessary	14070
Outlier/Noise	5793
Job Market Notification System	3741
Job Posting and Application Process	1728
Seasonal Offer Promotion Updates	1712
Advertisement Management Process	1593
Electronic Signature Orders	1298
Email Communication and Scheduling	596
Contact Issues and Rejection	581
Awaiting Response or Answer	441
Email Rejection and No Contact	321
Name: count, dtype: int64	

Note. Frequency counts of the 10 most common topic_action_label categories assigned to sales log messages. Labels were produced using the BERTopic pipeline with HDBSCAN clustering. The "Outlier/Noise" category captures messages that did not fit any coherent topic. "Follow-up Email Attempts Necessary" was the dominant class.

Figure B2

Comparison of Two Business-Aligned Category Labeling Schemes (Approach 1b)

<pre>1 df['category_label'].value_counts()</pre>	
category_label	
Streamlining Business Process Proposal Submission	4522
Optimize Invoice Status for Payment Processing	4353
Optimizing Business Strategy with Signed Proposal	4045
Optimize Quote Renewal Packages Received on Time	3715
Rejecting Rejected Orders for Registration Issue	3359
Streamlining Seasonal Job Offer Posting Process	2995
Optimizing Recruitment Post Moment Update Process	2979
Optimizing Job Market Offer Proposals	2418
Optimizing Seasonal Job Advertisements on JobsDB	1784
Optimizing Job Listings for Premium CV Posting	1704
Name: count, dtype: int64	
<pre>1 df['category_label_2'].value_counts()</pre>	
category_label_2	
Propose Signed Receipt for Posting	4522
Review and Revise Invoice Draft	4353
Propose Signed Logo Request	4045
Renewal Packages Received Reminder	3715
Investigate Rejected Order Issue	3359
Send Seasonal Job Offer Email Posting	2995
Update Recruitment Post Immediately	2979
Send Seasonal Job Offers via Email	2418
Optimize Job Advertisements for Seasonal Demand	1784
Propose Premium Job Listing Posting	1704
Name: count, dtype: int64	

Note. Both panels show frequency distribution of category labels. Both sets were derived from rule-based and LLM-refined post-processing of topic clusters.

Figure B3

Category Name Distribution from FASTopic (Approach 1c)

```
df['category_name'].value_counts()
```

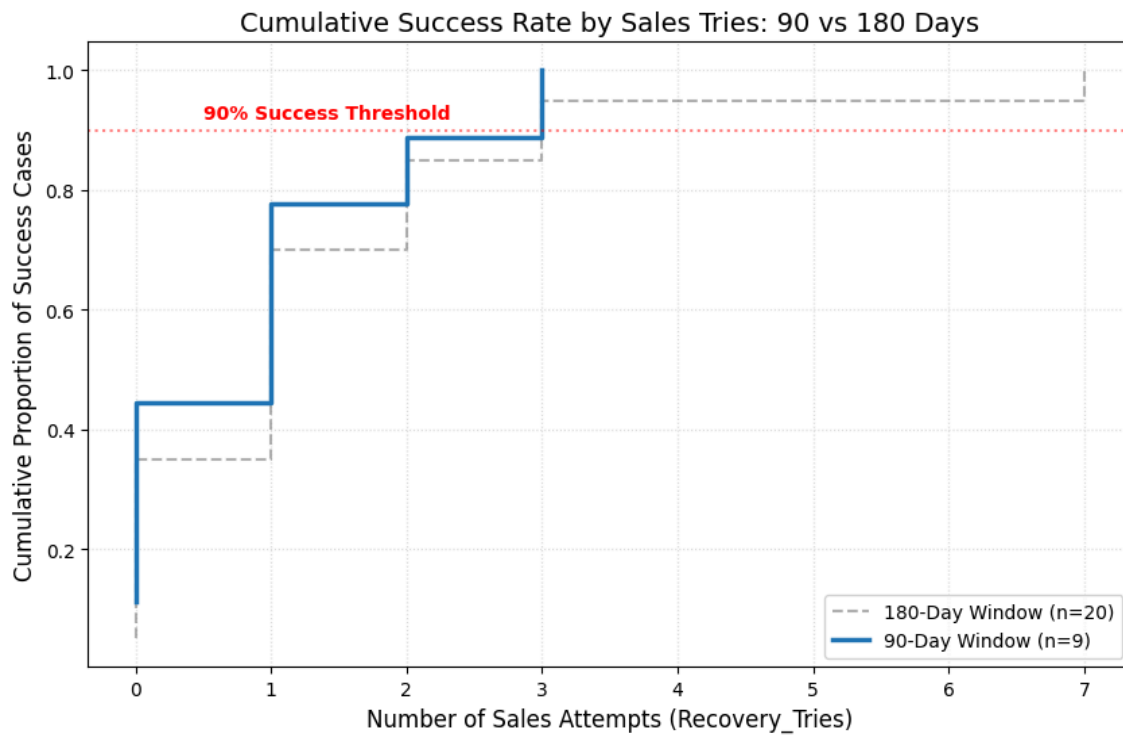
category_name	
Refining Job Specifications	12573
Verifying Candidate Qualifications	12365
Engaging with Clients for Feedback	3646
Executing Job Postings and Scheduling	1193
Configuring Client Accounts and Setup	685
Managing Sales Funnel Pipelines	486
Providing Platform Administrative Support	480
Processing Financial Reconciliations	364
Analyzing Performance Metrics	56
Ensuring Compliance with Regulations	26

Note. Bar chart of final activity category counts and percentages ($n \approx 31,874$ messages). Top categories: Refining Job Specifications (~39%), Verifying Candidate Qualifications (~39%). Derived from FASTopic clustering of ActionLog messages.

Appendix C

Figure C1

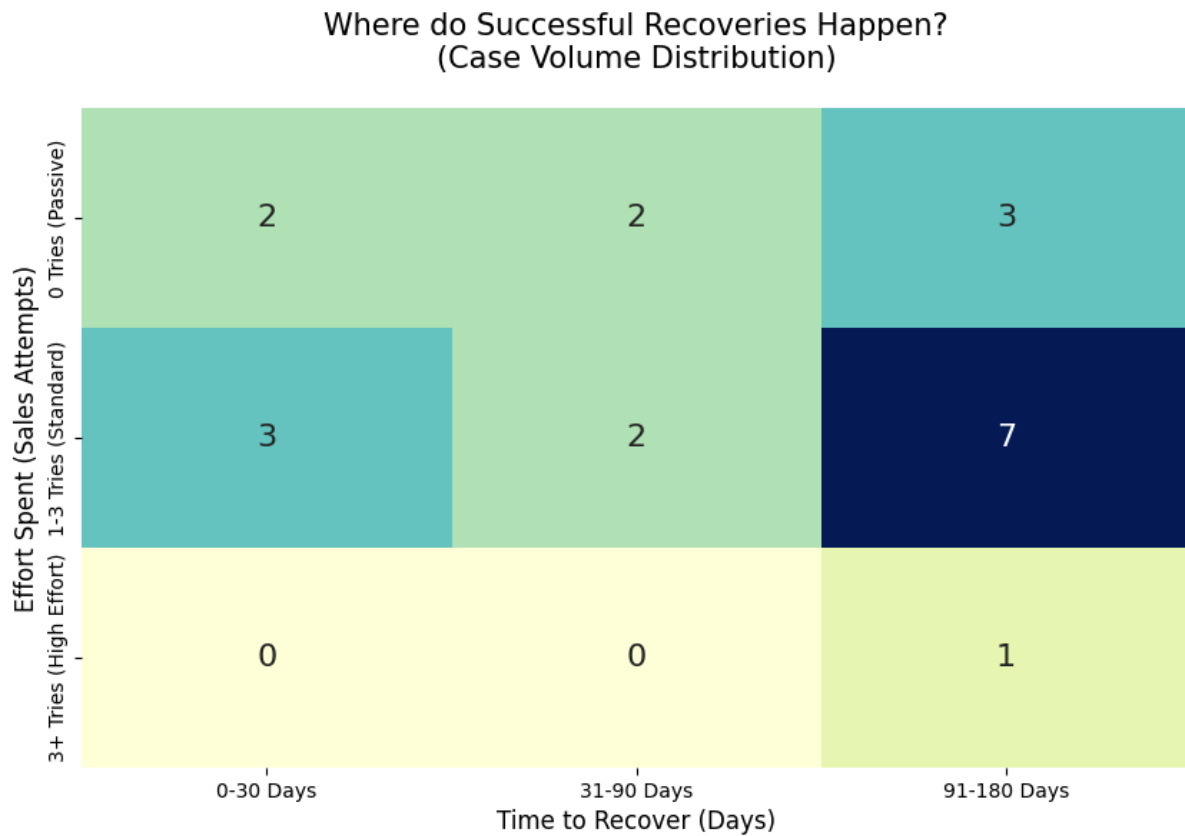
Cumulative Success Rate by Sales Attempts: 90-Day vs. 180-Day Window



Note. Step chart comparing cumulative proportion of successful recoveries by number of sales attempts for 20 cases. In 90-day window ($n = 9$), 100% of recoveries by 3 attempts; full 180-day window ($n = 20$) reaches ~90% by 3 attempts. Dashed line at 90% threshold.

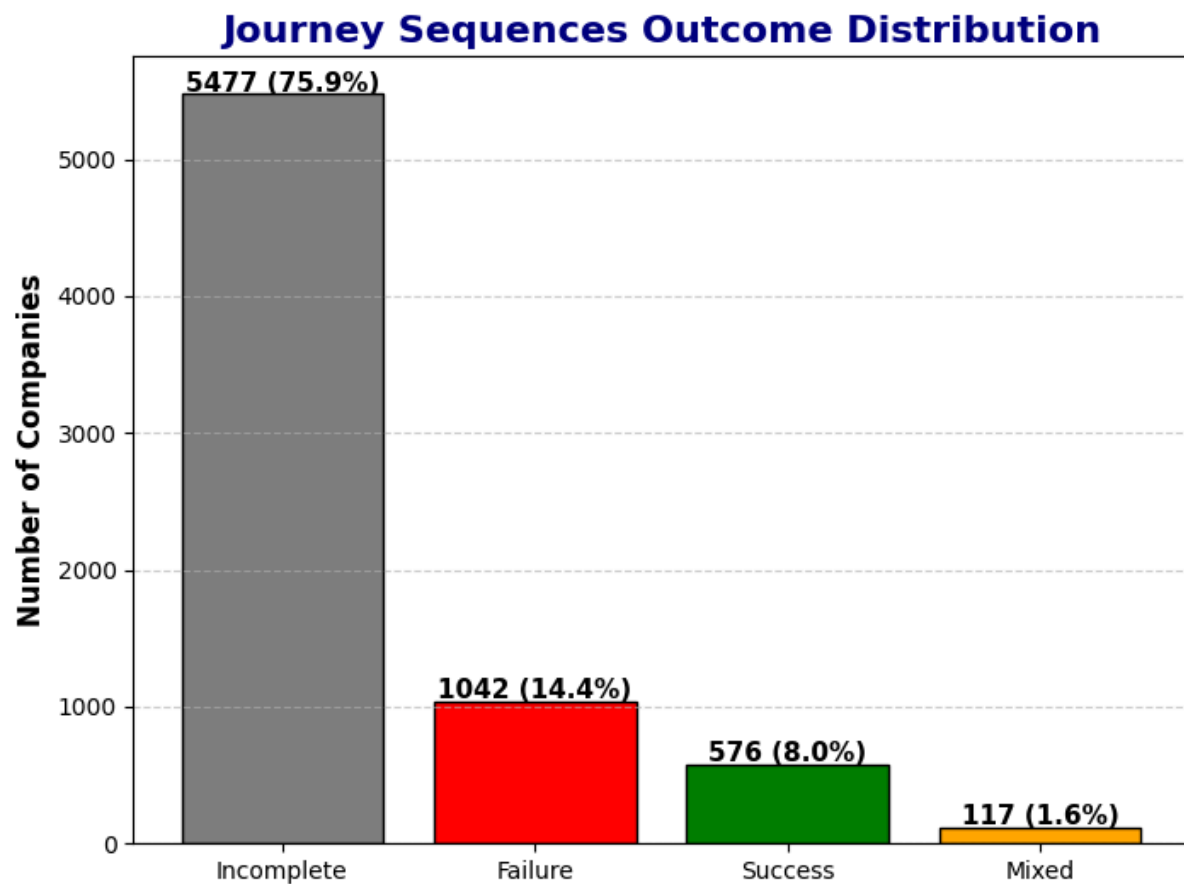
Figure C2

Where Successful Recoveries Happen? (Case Volume Distribution)



Note. Heatmap of 20 verified recovery cases (previously Rejected → later Signed) by time-to-recover (days) and sales effort level (attempts). Light = low/passive effort; blue = standard (1–3 attempts); dark = high effort (3+ attempts). Highest volume in late stage with standard effort.

Figure C3
Journey Sequences Outcome Distribution



Note. Bar chart showing distribution of 7,212 company-level sales journeys by outcome: Incomplete (75.9%, $n = 5,477$), Failure (14.4%, $n = 1,042$), Success (8.0%, $n = 576$), and Mixed (1.6%, $n = 117$). High incompletion rate indicates major opportunity for improved follow-up engagement.